Course: COM210 Engineering Maths III
Semester: 04/2023
Instructor: Abdul Basit

Quiz 2

Instructions:

- The due date is **28/06/23 at the start of class**. Alternatively you can email me a scan of your solutions before then. If you choose to do so, please make sure it is legible. Pictures will not be accepted.
- You should justify all your answers. An answer without justification will not receive any credit.

1. Find the linear approximation to $f(x, y) = \dfrac{10x^2}{x - y}$ at the point $(4, -1)$. Use this linear approximation to estimate the value of $(3.9, -0.8)$. [5]
2. Find the tangent plane to the surface given by $x^2y - 4ze^{x+y} + 35 = 0$ at the point $(3, -3, 2)$. [5]
3. Find and classify all critical points of $f(x, y) = 2x^3 + 2y^3 - 6xy$. [10]
4. Find the maximum and minimum values of $f(x, y) = 8xy + 1$ subject to the constraint $x^2 + 4y^2 = 8$. [10]
5. Evaluate the integral $\iint_D x(y - 1)\, dA$ where D is the region bounded by $y = 1 - x^2$ and $y = x^2 - 3$. [10]
6. Using polar coordinates, evaluate $\displaystyle\iint_D \frac{1}{1 + x^2 + y^2}\, dA$ where D is the region bounded by the circles $x^2 + y^2 = 1$, $x^2 + y^2 = 4$, and lying below the line $y = x$ and above the line $y = 0$. [10]